

Lab 3: Architectural Design Justification

SRN: PES1UG24CS584 | Project: Community Solar Credit Allocation Manager

Domain: Sustainability & Green Tech | Target Actors: Prosumer Resident, Co-op Manager

Architecture Selection: "We chose the Microservices Architecture for the Community Solar Credit Allocation Manager System."

1. Architectural Choice

The system is decomposed into six independently deployable microservices: a Resident & Co-op Manager Portal, Member & Ownership Management Service, Energy Generation Data Ingestion Service, Credit Allocation Engine, Billing & Statement Service, and Notification Service. Each service owns its own data and exposes well-defined provided/required interfaces, following the Microservices architectural style rather than a single layered monolith.

2. Two Scenario-Related Reasons

Independent scaling for variable workloads: Smart-meter and inverter readings stream continuously into the Energy Generation Data Ingestion Service throughout the day, while resident portal traffic and credit-allocation runs peak around month-end billing. Microservices allow the ingestion pipeline and the Credit Allocation Engine to be scaled independently of low-traffic services such as Notifications, avoiding wasted infrastructure spend.

Fault isolation as the co-operative grows: As more prosumer residents and additional solar arrays join the co-op, a failure in one service does not cascade to the rest of the system. For example, if the Notification Service is temporarily down, residents can still log into the Portal and view accurate credit balances and statements, because Credit Allocation and Billing remain unaffected.

3. Security Advantage

Because each microservice owns its own data store and communicates only through explicit interfaces (e.g., MemberShareInterface, CreditAllocationInterface), sensitive ownership and financial data held by the Member & Ownership Management Service and the Billing & Statement Service is isolated behind service boundaries. Each service can enforce its own authentication and access control, so a compromise of a lower-trust, public-facing component such as the Notification Service does not directly expose the credit ledger or billing records — the blast radius of any single breach is contained.

4. Performance Benefit

Decoupling the Energy Generation Data Ingestion Service from the Credit Allocation Engine lets high-frequency meter readings be buffered and processed asynchronously, so continuous data ingestion never blocks resident-facing Portal requests. The Credit Allocation Engine can likewise be scaled out or run in scheduled batches during month-end processing without slowing down everyday portal browsing, statement viewing, or notification delivery.